

Smart Meter Tamper Detection

This folder contains a complete smart-meter tamper-detection workflow. It combines data cleaning, rule-based labeling, sequence models, feeder-level graph features, late fusion, and evaluation artifacts.

Workflow

Run the notebooks in this order when rebuilding the project:

1. ``data_cleaning.ipynb`` reads the four raw Excel files, regularizes each meter's timestamps, removes duplicates, handles missing timestamps, and writes ``Dataset/cleaned_data.csv``.
2. ``data_labeling.ipynb`` reads the cleaned CSV, applies the documented electrical tamper rules and missing-value/power-cut overrides, and writes the labeled dataset. The notebook currently saves ``clean_data_labeled.csv`` in its working directory; the checked-in training notebooks expect ``Dataset/clean_data_labeled.csv``.
3. ``hybrid_lstm_train.ipynb`` trains the sequence and feeder-graph branches, extracts embeddings, trains the late-fusion classifier, evaluates it, and saves the LSTM-family model artifacts.
4. ``hybrid_lstm_evaluate.ipynb`` loads the saved LSTM-family artifacts and produces test predictions, metrics, confusion matrices, PR curves, ROC curves, and summary charts.
5. ``hybrid_bilstm_train.ipynb`` trains the corresponding BiLSTM-family hybrid model using a shared global time cutoff and severe-dropout handling.
6. ``hybrid_bilstm_evaluate.ipynb`` evaluates the BiLSTM-family artifacts without retraining or modifying the labeled dataset.

The model configuration uses 48-reading causal windows, a ResNet1D feature extractor, an LSTM classifier, a star-shaped feeder graph with one message-passing layer, and a final late-fusion classifier. The saved run configuration records the 80% global training cutoff, feature columns, target columns, and model dimensions.

Directory and file reference

Source data and datasets

- ``Dataset/1.xlsx`` - Raw smart-meter workbook 1 used by the cleaning notebook.
- ``Dataset/2.xlsx`` - Raw smart-meter workbook 2 used by the cleaning notebook.
- ``Dataset/3.xlsx`` - Raw smart-meter workbook 3 used by the cleaning notebook.
- ``Dataset/4.xlsx`` - Raw smart-meter workbook 4 used by the cleaning notebook.
- ``Dataset/Tamper Thresholds (10-60A).doc`` - Reference document for the electrical tamper thresholds and occurrence.
- ``Dataset/cleaned_data.csv`` - Cleaned, timestamp-regularized, unlabeled meter readings.
- ``Dataset/clean_data_labeled.csv`` - Cleaned readings with per-tamper binary flags and the ``Tamper_Labels`` metadata.
- ``Dataset/fusion_evaluation_metrics.csv`` - Evaluation metrics for a fusion run stored with the dataset outputs.

Notebooks

- ``data_cleaning.ipynb`` - Loads the four Excel workbooks, standardizes meter histories, handles duplicates and missing timestamps, and writes the cleaned CSV.
- ``data_labeling.ipynb`` - Applies rule-based labels including current/voltage imbalance, voltage and current thresholds, and power-cut overrides.
- ``hybrid_lstm_train.ipynb`` - Trains the LSTM-family ResNet1D plus LSTM branch, feeder GNN branch, and late-fusion classifier.
- ``hybrid_lstm_evaluate.ipynb`` - Reconstructs the test set and evaluates the saved LSTM-family hybrid model.

- `hybrid_bilstm_train.ipynb` - Trains the BiLSTM-family hybrid model with a shared train/test cutoff and masked labels.
- `hybrid_bilstm_evaluate.ipynb` - Evaluates the saved BiLSTM-family hybrid model and generates detailed metrics.

`lstm_models/`

Saved model, preprocessing, configuration, and training-evaluation artifacts for the LSTM-family run:

- `resnet1d_extractor.pt` - Learned ResNet1D sequence feature-extractor weights.
- `lstm_classifier.pt` - Learned LSTM classifier weights.
- `feeder_gnn.pt` - Learned feeder graph/message-passing weights.
- `fusion_classifier.pt` - Learned classifier over the sequence and feeder embeddings.
- `seq_scaler.joblib` - Fitted scaler for sequential features.
- `tab_scaler.joblib` - Fitted scaler for rolling/tabular features.
- `run_config.json` - Model dimensions, feature names, target names, split cutoff, and window configuration.

`bilstm_models/`

Saved model, preprocessing, configuration, and training-evaluation artifacts for the BiLSTM-family run:

- `resnet1d_extractor.pt` - Learned ResNet1D sequence feature-extractor weights.
- `lstm_classifier.pt` - Learned recurrent classifier weights used by this saved run.
- `feeder_gnn.pt` - Learned feeder graph/message-passing weights.
- `fusion_classifier.pt` - Learned classifier over the sequence and feeder embeddings.
- `seq_scaler.joblib` - Fitted scaler for sequential features.
- `tab_scaler.joblib` - Fitted scaler for rolling/tabular features.
- `run_config.json` - Model dimensions, feature names, target names, split cutoff, and window configuration.
- `fusion_evaluation_metrics.csv` - Per-tamper metrics from the fusion evaluation performed during training.
- `positive_only_breakdown.csv` - True-positive and false-negative counts restricted to actual positive events.

`lstm_evaluate/`

Evaluation results generated from the LSTM-family saved model:

- `evaluation_predictions.csv` - Long-form test predictions, true labels, predicted labels, and probabilities.
- `evaluation_results.csv` - Per-tamper confusion counts and metrics including Accuracy, Precision, Recall, F1, Score, and AUC.
- `positive_only_breakdown.csv` - Per-tamper positive-event totals, detected positives, missed positives, and recall.
- `confusion_matrices.npz` - Raw confusion-matrix arrays.
- `confusion_matrices.png` - Rendered confusion-matrix grid.
- `pr_curve_data.npz` - Raw precision-recall curve points.
- `pr_curves.png` - Precision-recall curve plots.
- `roc_curve_data.npz` - Raw false-positive-rate/true-positive-rate curve points.
- `roc_curves.png` - ROC curve plot. The filename currently begins with a space.
- `all_metrics_summary.png` - Chart comparing scalar metrics across tamper types.

`bilstm_evaluate/`

Evaluation results generated from the BiLSTM-family saved model. The files have the same

roles as the corresponding LSTM evaluation files:

- `evaluation\_predictions.csv` - Test predictions, labels, and probabilities.
- `evaluation\_results.csv` - Per-tamper confusion counts and aggregate metrics.
- `positive\_only\_breakdown.csv` - Positive-event detection breakdown.
- `confusion\_matrices.npz` and `confusion\_matrices.png` - Raw and visual confusion matrices.
- `pr\_curve\_data.npz` and `pr\_curves.png` - Raw and visual precision-recall curves.
- `roc\_curve\_data.npz` and `roc\_curves.png` - Raw and visual ROC curves.
- `all\_metrics\_summary.png` - Summary chart of scalar metrics.

Other file

- `.DS\_Store` - macOS Finder metadata; it is not part of the data or model pipeline.

Notes for rerunning

- Use a Python environment with Jupyter, pandas, NumPy, PyTorch, scikit-learn, joblib, matplotlib, and openpyxl.
- Run notebooks from the project root so relative paths such as `Dataset/...` resolve correctly.
- The training and evaluation notebooks use `Dataset/clean\_data\_labeled.csv` as read-only input.
- The notebook code contains historical output-path references (`model/` and `bilstm\_evaluate1/`), while the artifacts are in the current directory.
- Model performance should be judged primarily with Recall, Precision, F1, and FPR because tamper events are rare.